Liquid Cooling Approaches for E1.S SSDs

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Agenda



- ❑ Thermal challenge for E1.S SSDs
- ❑ Why direct liquid cooling (DLC) matters
- ❑ KIOXIA NX1 Series overview
- ❑ Cold-plate DLC design
- ❑ Performance benefits
- ❑ Industry collaboration and ecosystem alignment
- ❑ Summary

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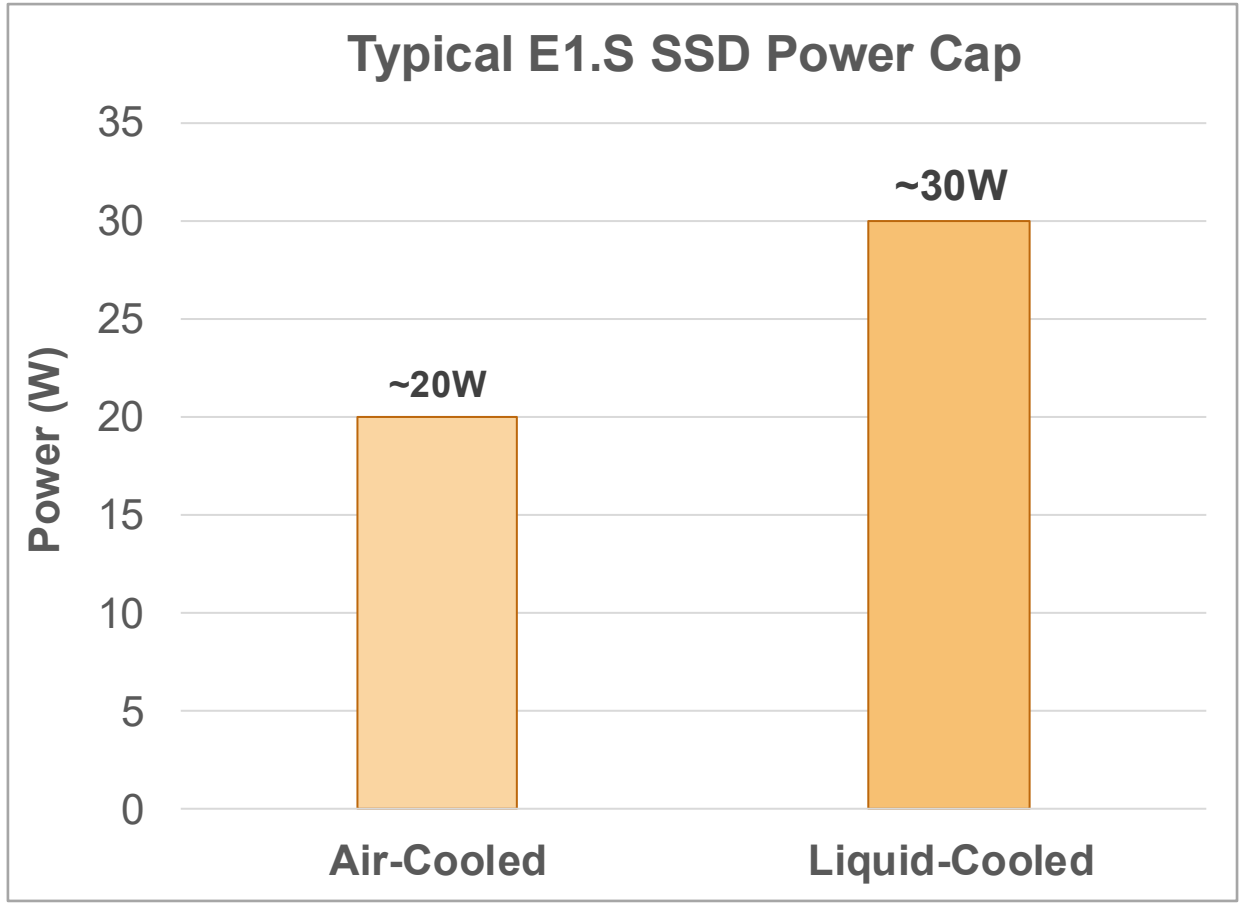
Thermal Challenge for E1.S SSDs

Higher Performance and Density Drive Higher Power and Heat

E1.S SSDs in AI and hyperscale systems operate under heavy sustained workloads

Air-cooled SSD design is less effective at removing heat in a higher power environment leading to thermal throttling

Thermal throttling impacts latency, quality of service, and overall system performance



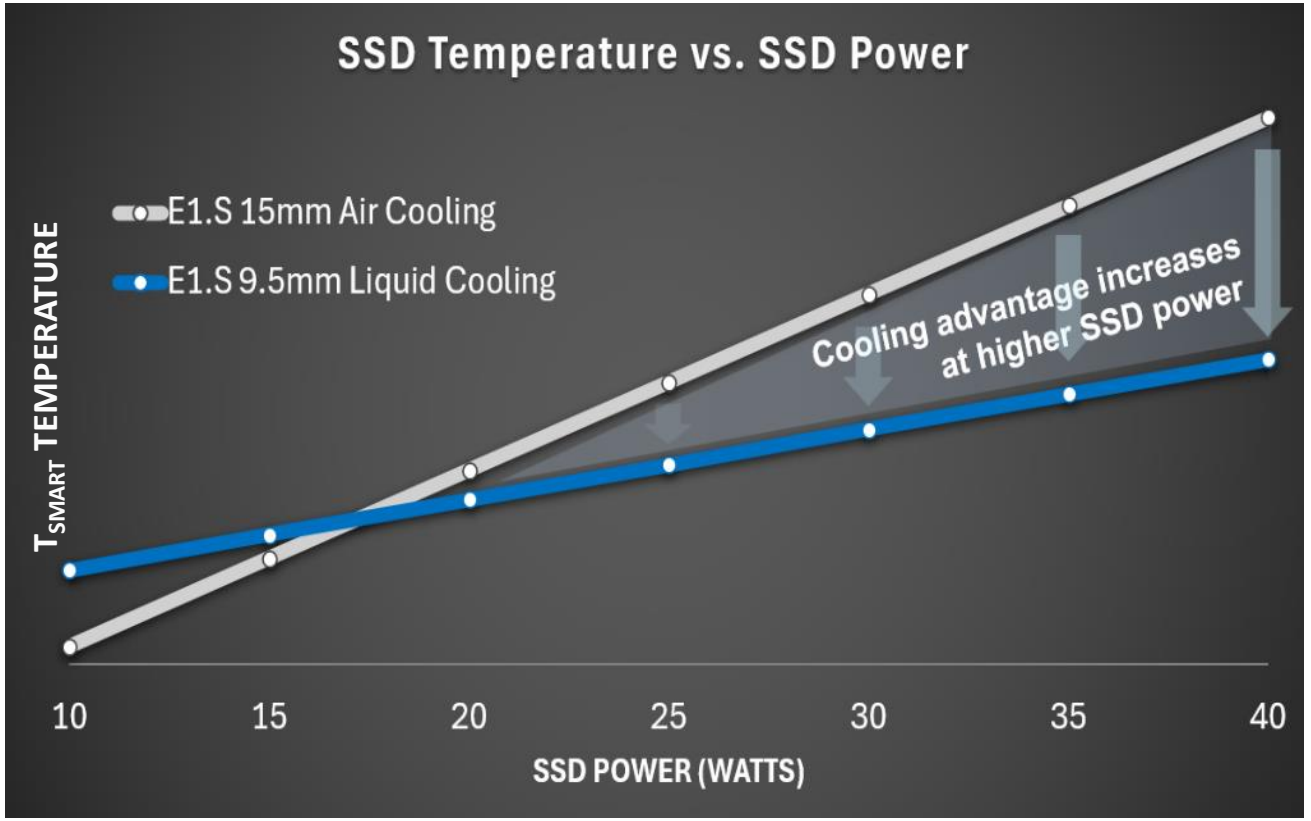
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Why Liquid Cooling Matters

Cold-Plate Direct Liquid Cooling Maintains Stable Thermal Performance

- Efficient heat removal keeps SSD temperature lower and consistent
- Can sustain higher performance under heavy workloads, compared to air cooled
- Enable higher power envelopes to maximize performance
- Can improve reliability and longevity with predictable environmental temperature, compared to air cooled

Liquid cooling helps unlock higher SSD power envelopes while maintaining operational temperature.[1]



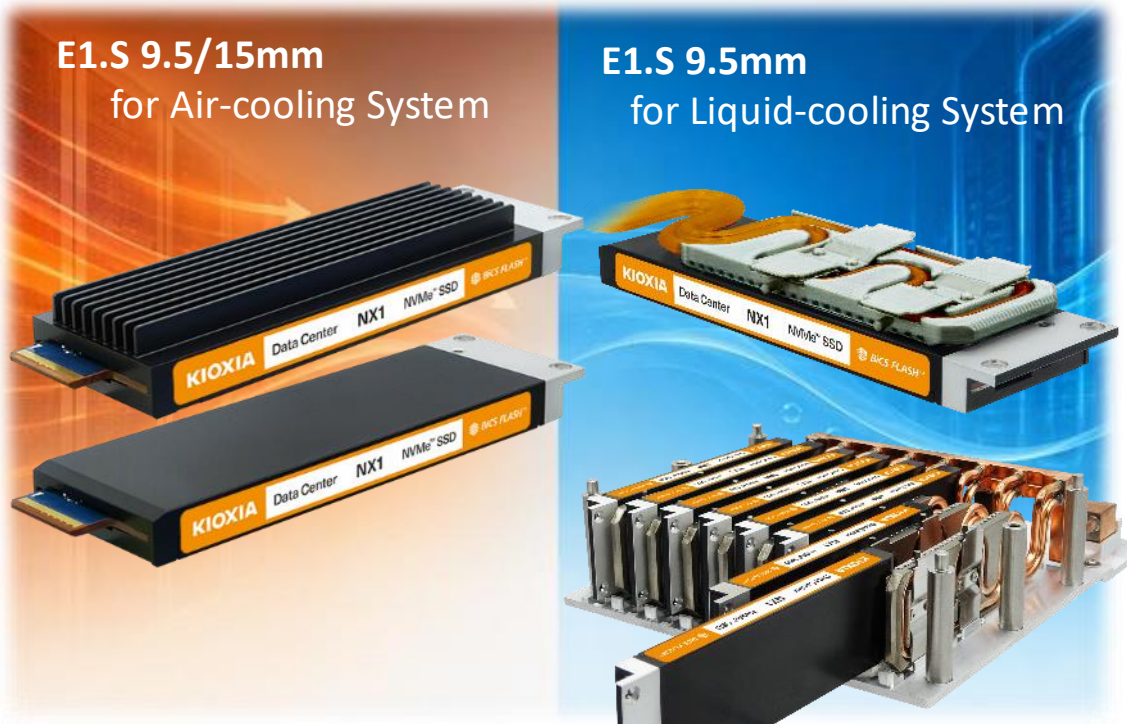
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1. Maintaining operational temperature to prevent SSD performance throttling.

KIOXIA NX1 Series E1.S SSD Overview

Key Features

- ❑ KIOXIA BiCS FLASH™ generation 8 TLC flash memory
- ❑ PCIe® 5.0 x4 interface, NVMe™ 2.0e specifications
- ❑ OCP® Datacenter NVMe SSD Specification 2.6 support
- ❑ Capacities[2] up to 15.36 terabytes (TB)
- ❑ EDSFF E1.S form factor (9.5mm* and 15mm)
- ❑ Designed for hyperscale and AI applications
- ❑ 1 drive write per day (DWPD) endurance
- ❑ Advanced power loss protection and data path protection



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KIOXIA NX1 Series E1.S SSD with Cooler Master® cold plate assembly

* Only the 9.5 mm option supports cold-plate direct liquid cooling (DLC)

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2. Definition of capacity: KIOXIA Corporation defines a megabyte (MB) as 1,000,000 bytes, a gigabyte (GB) as 1,000,000,000 bytes, a terabyte (TB) as 1,000,000,000,000 bytes and a petabyte (PB) as 1,000,000,000,000,000 bytes. A computer operating system, however, reports storage capacity using powers of 2 for the definition of 1Gbit = 2^{30} bits = 1,073,741,824 bits, 1GB = 2^{30} bytes = 1,073,741,824 bytes, 1TB = 2^{40} bytes = 1,099,511,627,776 bytes and 1PB = 2^{40} bytes = 1,125,899,906,842,624 bytes and therefore shows less storage capacity. Available storage capacity (including examples of various media files) will vary based on file size, formatting, settings, software and operating system, and/or pre-installed software applications, or media content. Actual formatted capacity may vary.

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SNIA® SFF-TA-1006 Rev. 2.0 Compliance

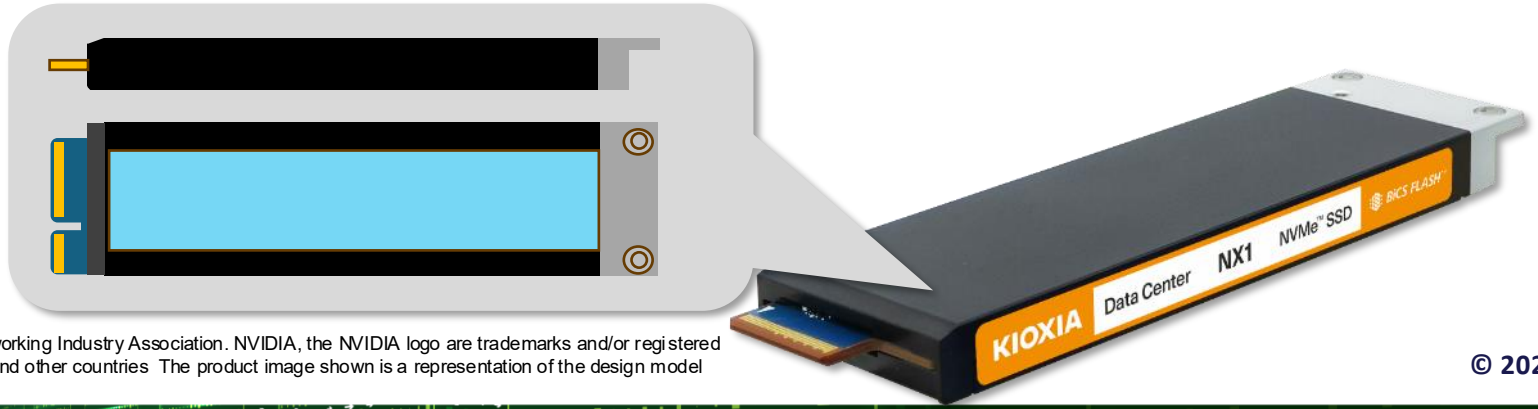
KIOXIA NX1 Series E1.S SSD is Designed to the SFF-TA-1006 Rev. 2.0 Specification to Support Direct Cold-Plate Cooling

SFF-TA-1006 Rev. 2.0 Highlights

- ❑ Defines E1.S mechanical design to accommodate thermal cold-plate direct liquid cooling
- ❑ Standardizes dimensions
- ❑ Enables ecosystem adoption and interoperability

KIOXIA Compliance

- ❑ Exterior design to comply with standards
- ❑ Validates with reference cold plate solutions
- ❑ Supports NVIDIA reference design platforms
- ❑ Single SSD design supports both air-cooled and liquid-cooled systems



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Industry Collaboration and Ecosystem Alignment

OCP® Collaboration

- Active contributor to the OCP Storage Project
- Support liquid cooling innovation for next-generation data centers
- Align with community specifications and best practices

SNIA® Collaboration

- Align with SNIA SFF standards for E1.S liquid cooling solutions
- Promote interoperability across the ecosystem

NVIDIA Reference Design Alignment

- KIOXIA NX1 Series E1.S SSD is designed to support NVIDIA reference design platforms for liquid-cooled storage
- Deliver compatibility for AI infrastructure
- Develop E3.S direct liquid cooled storage solution to support NVIDIA® CMX™ context memory storage platform.

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Summary

- Thermal management is critical for high-performance E1.S applications in AI and hyperscale deployments
- Cold-plate direct liquid cooling (DLC) can effectively lower temperatures and maintain stable performance
- KIOXIA NX1 Series E1.S SSD is designed to support DLC and compliant with SNIA® SFF-TA-1006 Rev 2.0 and OCP® specifications
- Collaboration with NVIDIA for next-generation liquid cooling platforms
- Liquid cooling enables a higher power ceiling[2] and optimizes SSD efficiency for the AI era

KIOXIA NX1 Series E1.S SSD



Engineered for High Density Applications, Efficiency, and Reliability.

Ready for Direct Liquid Cooling

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2. Compared to air cooled.

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Call to Action

Stop by the KIOXIA booth, **F9**, to connect with our team and explore how our solutions can support your specific requirements.



Near the Expo Hall Stage.



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Thank You!

